Quantification: 872_wt_hp_z06
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	14
interior lacuna count	13
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 13)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	1480.77	1363	701.374	px^2
major_axis_length_px	lacuna shape	82.2465	82.9994	24.4938	px
minor_axis_length_px	lacuna shape	25.3423	22.6854	8.2386	px
aspect_ratio	lacuna shape	3.54	3.4602	1.6626	unitless
eccentricity	lacuna shape	0.9286	0.9573	0.0748	unitless
solidity	lacuna shape	0.8185	0.8709	0.1073	unitless
roots_count	headline	10.3846	10	4.942	count
ring_length_r30_px	headline	442.846	396	132.762	px
ring_length_r60_px	reported	1086.85	1031	378.559	px
owned_length_px	ownership-dependent	2533.42	2506.47	1284.81	px
edge_count	ownership-dependent	159.615	151	87.4343	count
mean_edge_length_px	ownership-dependent	16.0572	15.9401	1.2042	px
perimeter_px	normalised	212.149	215.095	60.0168	px
ring_area_r30_px2	normalised	8168.77	8557	1261.94	px^2
ring_area_r60_px2	normalised	20050.1	20339	2795.1	px^2
in_frame_fraction_r30	normalised	0.9918	1	0.0297	unitless
in_frame_fraction_r60	normalised	0.9854	1	0.0528	unitless
ring_density_r30	normalised	0.0542535	0.0530547	0.0135123	px^-1
ring_density_r60	normalised	0.0538827	0.0547778	0.0166206	px^-1
roots_per_100px_perimeter	normalised	4.9458	5.4831	2.0694	per 100 px
ring_attached_length_r30_px	network v2	415.154	370	131.677	px
ring_attached_length_r60_px	network v2	956.385	817	410.914	px
ring_length_w_r30_px	network v2	503.861	444.132	153.19	px
ring_length_w_r60_px	network v2	1238.45	1160.82	431.632	px
sholl_crossings_r10	network v2	9.6154	10	3.5949	count
sholl_crossings_r20	network v2	12.4615	12	4.3706	count
sholl_crossings_r30	network v2	14.2308	15	4.7285	count
loop_count	loops, report only	22.4615	19	17.2753	count
loop_edge_share	loops, report only	0.4683	0.493	0.1771	share
ring_width_mean_r30_px	width	7.403	7.5301	0.5746	px

Per field
----

measure	value	units
analysed_area_px2	776953	px^2
total_skeleton_length_px	38138	px
canalicular_length_density_per_px	0.0490866	px^-1
junction_count	1271	count
junction_density_per_px2	0.00163588	px^-2
lacuna_count	14	count

lacunae_per_px2	1.802e-05	px^-2
skeleton_component_count	159	count
mean_component_length_px	239.862	px
median_component_length_px	17	px
field_length_density_w_per_px	0.0557092	px^-1
field_density_without_flagged_per_px	0.038163	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	23	px
unattached_component_count	2	count
canalicular_length_density_attached_per_px	0.0490866	px^-1
field_length_density_with_unattached_per_px	0.0491162	px^-1
excluded_area_px2	251704	px^2
analysed_field_area_px2	796872	px^2
loop_count	400	count
loop_edge_share	0.5428	share
edge_count	2710	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.5875	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	38952	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1057	102.462	18.2811	5.6048	0.984	0.7152	-1.5145	124.814	413.139	False
2	1545	102.844	28.21	3.6457	0.9616	0.5974	-1.5363	130.782	502.108	False
3	2178	97.2133	31.5882	3.0775	0.9457	0.7666	-1.4507	132.953	694.614	False
4	1024	110.295	13.9978	7.8795	0.9919	0.6469	1.515	141.115	876.421	False
5	1532	90.3989	22.6854	3.9849	0.968	0.8855	-1.4786	133.692	263.431	False
6	1458	82.9994	23.987	3.4602	0.9573	0.8423	-1.4301	272.217	418.857	False
7	2267	64.9571	46.3919	1.4002	0.6999	0.896	-1.1815	380.946	282.045	False
8	3165	121.397	33.5624	3.6171	0.961	0.8709	-1.4322	386.945	67.1896	False
9	743	47.9512	20.7766	2.3079	0.9013	0.8888	-1.5064	382.218	839.734	False
10	712	46.6981	20.2555	2.3054	0.901	0.8856	-1.5028	499.643	510.035	False
11	1363	80.3055	21.9882	3.6522	0.9618	0.8961	-1.3459	561.869	697.237	False
12	669	39.9711	24.8769	1.6068	0.7827	0.7707	-1.1357	585.831	14.0269	True
13	1360	67.805	25.7054	2.6378	0.9254	0.9484	-1.4973	614.482	323.551	False
14	846	53.8777	22.02	2.4468	0.9127	0.8004	1.5387	642.791	211.617	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	244	442	457.191	28	16.3282	7.5301
2	8	373	713	991.607	67	14.8001	7.887
3	10	465	884	1747.05	122	14.3201	7.8301

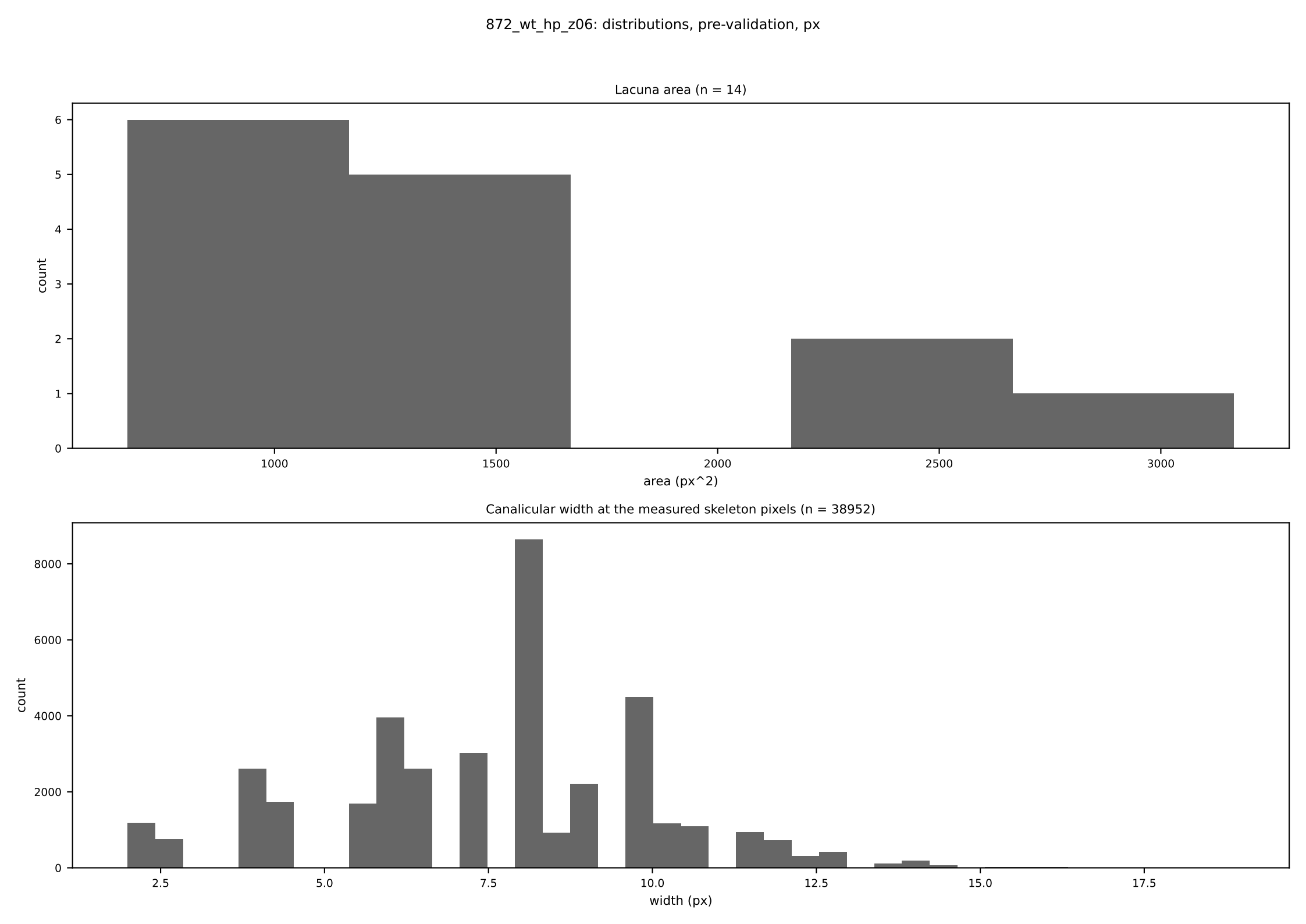
4	12	379	965	2499.16	151	16.5507	7.7983
5	12	349	696	2531.05	159	15.9186	6.7785
6	20	568	1570	3235.21	206	15.7049	7.7049
7	13	578	1524	2231.61	140	15.9401	8.0859
8	18	741	1762	5347.95	367	14.5721	7.3404
9	8	370	1051	3042.28	173	17.5854	7.2474
10	4	309	986	1540.44	89	17.3083	6.4548
11	12	476	1200	2614.61	142	18.4127	6.958
12	8	353	919	2462.03	161	15.2921	7.9105
13	3	396	1031	2506.47	156	16.0671	6.5111
14	9	509	1305	4189.88	275	15.2359	8.1131

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	231.817	7355	14578	1	1	0.0331747	0.0303197	2.5882
2	285.137	8557	19005	1	1	0.04359	0.0375164	2.8057
3	292.451	10015	23365	1	1	0.0464304	0.0378344	3.4194
4	241.681	9364	22992	1	1	0.0404742	0.0419711	4.9652
5	218.853	8669	20339	1	1	0.0402584	0.03422	5.4831
6	215.095	8592	22200	1	1	0.066108	0.0707207	9.2982
7	212.409	8564	22184	1	1	0.0674918	0.0686982	6.1203
8	301.179	9919	22490	0.893	0.8096	0.0747051	0.0783459	5.9765
9	123.154	6283	18209	1	1	0.0588891	0.0577187	6.4959
10	119.983	6180	18000	1	1	0.05	0.0547778	3.3338
11	196.024	8388	22420	1	1	0.0567477	0.0535236	6.1217
12	129.29	4152	11045	0.6578	0.6072	0.0850193	0.0832051	6.1876
13	167.539	7464	18130	1	1	0.0530547	0.0568671	1.7906
14	152.61	6844	16739	1	1	0.0743717	0.0779616	5.8974

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	236	413	279.551	503.576	6	8	5
2	370	667	444.132	831.139	6	8	10
3	456	850	528.262	994.654	10	12	10
4	352	965	420.037	1083.83	8	12	14
5	347	602	401.635	796.713	7	12	10
6	562	1549	639.203	1779.84	13	18	16
7	531	1420	661.973	1752.52	11	16	21
8	702	1735	850.754	2008.66	19	22	21
9	330	745	416.806	1219.37	10	9	16
10	244	584	340.208	1117.06	6	7	15
11	436	817	543.961	1357.57	11	15	19
12	347	887	409.806	1051.35	6	11	10
13	347	798	441.463	1160.82	8	10	12
14	484	1288	582.203	1494.06	10	13	16


872_wt_hp_z06: distributions, pre-validation, px
Lacuna area (n = 14)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 38952)
count
width (px)